# DEEP LEARNING-BASED HUMAN FACE AUTHENTICITY DETECTION

**MILESTONE 1**

AN EXPLAINABLE DUAL-STREAM VISION TRANSFORMER WITH RGB–FREQUENCY FUSION FOR ROBUST, TRANSPARENT DEEPFAKE IMAGE DETECTION.

# TEAM MEMBERS

- Rohit - Project Objectives & Problem Definition Lead
- Raunak - Literature Review & Benchmark Analysis Lead
- Vishakha - Research Findings & Comparative Analysis Lead
- Aman - Baseline Performance & Evaluation Strategy Lead
- Somendu - Research, data collection and processing Lead

# PROBLEM STATEMENT

AI-generated faces created using GANs and Diffusion Models are increasingly used for misinformation, identity theft, and digital fraud. Existing detectors struggle with unseen manipulations, compressed images, and explainability, highlighting the need for a robust and transparent deepfake detection framework.

**Rapid Spread**
Deepfake content proliferates across social media, news platforms, and digital channels, outpacing detection.

**300% Annual Growth**
Deepfake datasets expand by ~300% per year, demanding continuously evolving detection methods.

**Detection Gap**
Existing CNN-based detectors struggle with compressed media, unseen generators, and cross-dataset scenarios.

# FIVE CRITICAL LIMITATIONS

- RGB-Only Feature Learning
  Most existing detectors rely only on spatial RGB features, missing hidden frequency-domain artifacts.
- Poor Cross-Dataset Generalization
  Models often overfit to training datasets and perform poorly on unseen manipulation techniques.
- Sensitivity to Compression & Post-Processing
  Detection accuracy decreases when images are compressed, resized, or captured through screenshots/camera recapture.
- Limited Explainability
  Most existing systems act as black boxes and do not explain which facial regions influenced the prediction.
- No Automated Forensic Reporting
  Current approaches lack interpretable reports containing confidence scores, visual explanations, and detection details.

# PROJECT OBJECTIVES

**Explainable ViT Framework**
Develop a Dual-Stream Vision Transformer with RGB–frequency fusion for accurate real vs. deepfake classification.

**Cross-Attention Fusion**
Combine RGB spatial features with FFT/DCT frequency analysis via cross-attention to capture complementary artifacts.

**Cross-Dataset Generalization**
Evaluate across FaceForensics++, Celeb-DF, and DFDC to assess robustness against unseen generators.

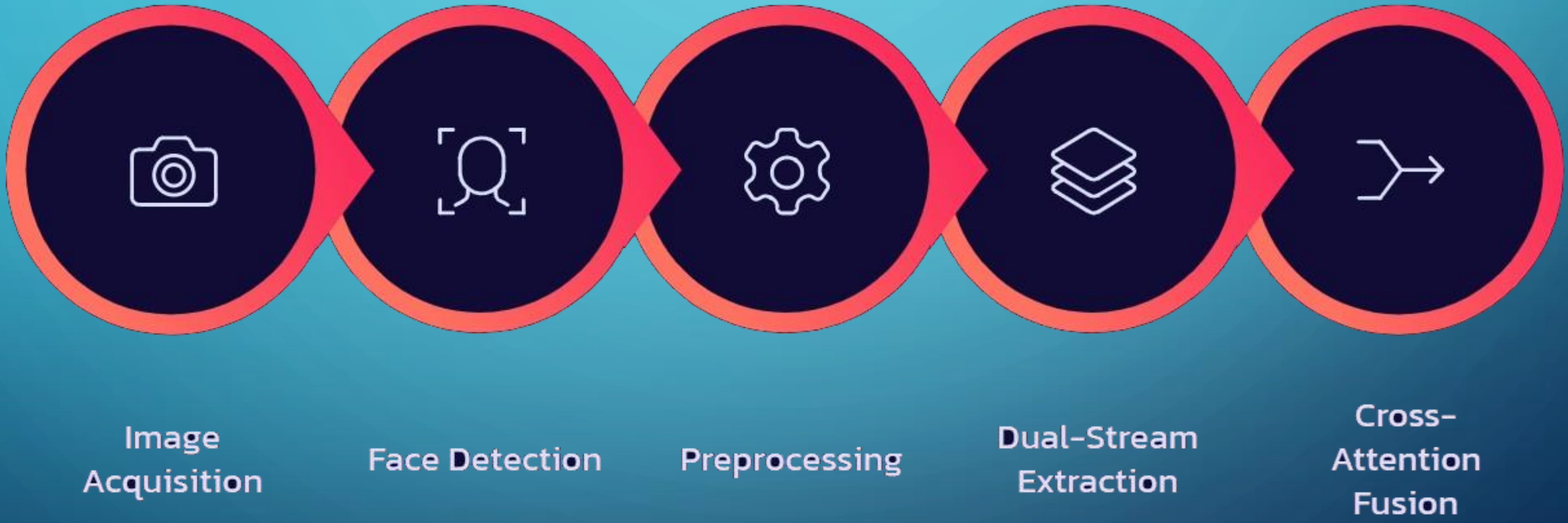
**Transparent Predictions**
Use Attention Rollout and frequency saliency maps to highlight influential facial regions and spectral patterns.

# HOW DEEPFAKES HAVE EVOLVED

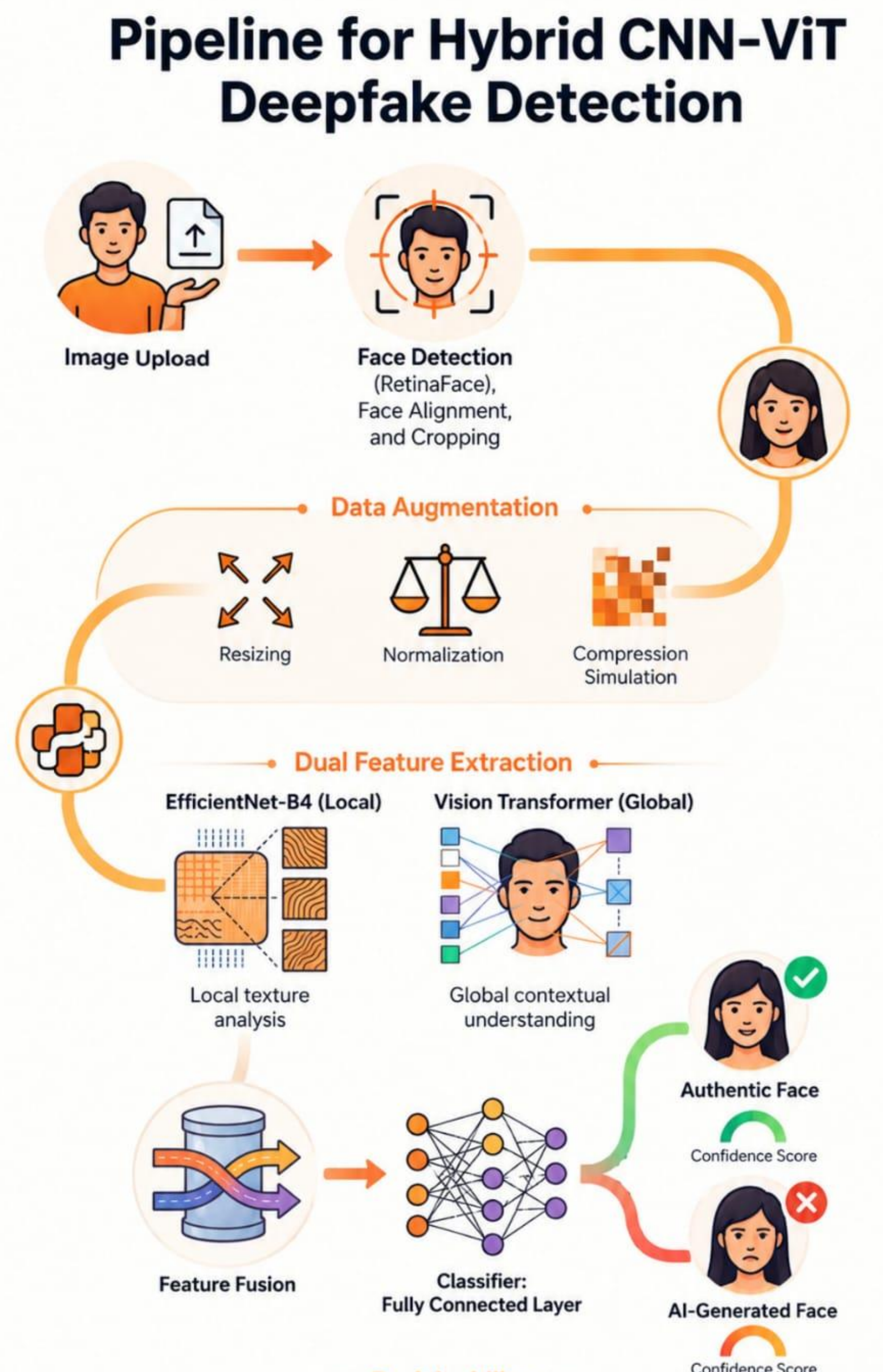
| Feature | Early Generation (~2018) | Modern SOTA |
| --- | --- | --- |
| **Facial Details** | Blurry ears, mismatched eyes, distorted teeth | Realistic skin texture, consistent features, natural lighting |
| **Geometry & Physics** | Incorrect shadows, hair artifacts, inconsistent backgrounds | Accurate geometry, realistic reflections and shadows |
| **Detection Difficulty** | Easily detected using conventional CNNs | Requires advanced deep learning to detect subtle artifacts |

Diffusion-based generators have dramatically narrowed the gap between synthetic and authentic facial imagery, rendering traditional CNN-based detectors increasingly ineffective.

# EIGHT-STAGE DETECTION PIPELINE



# ARCHITECTURE



# EXPLAINABILITY

- Why Explainability?

Increases **transparency** and **user trust.**

Explains **why** an image is classified as Real or Deepfake.

Helps identify manipulated facial regions and hidden artifacts.

**Attention Rollout (ViT)**

- Highlights facial regions influencing the prediction.
- Focuses on **eyes, mouth, hairline, and jawline.**

**Frequency-Domain Visualization**

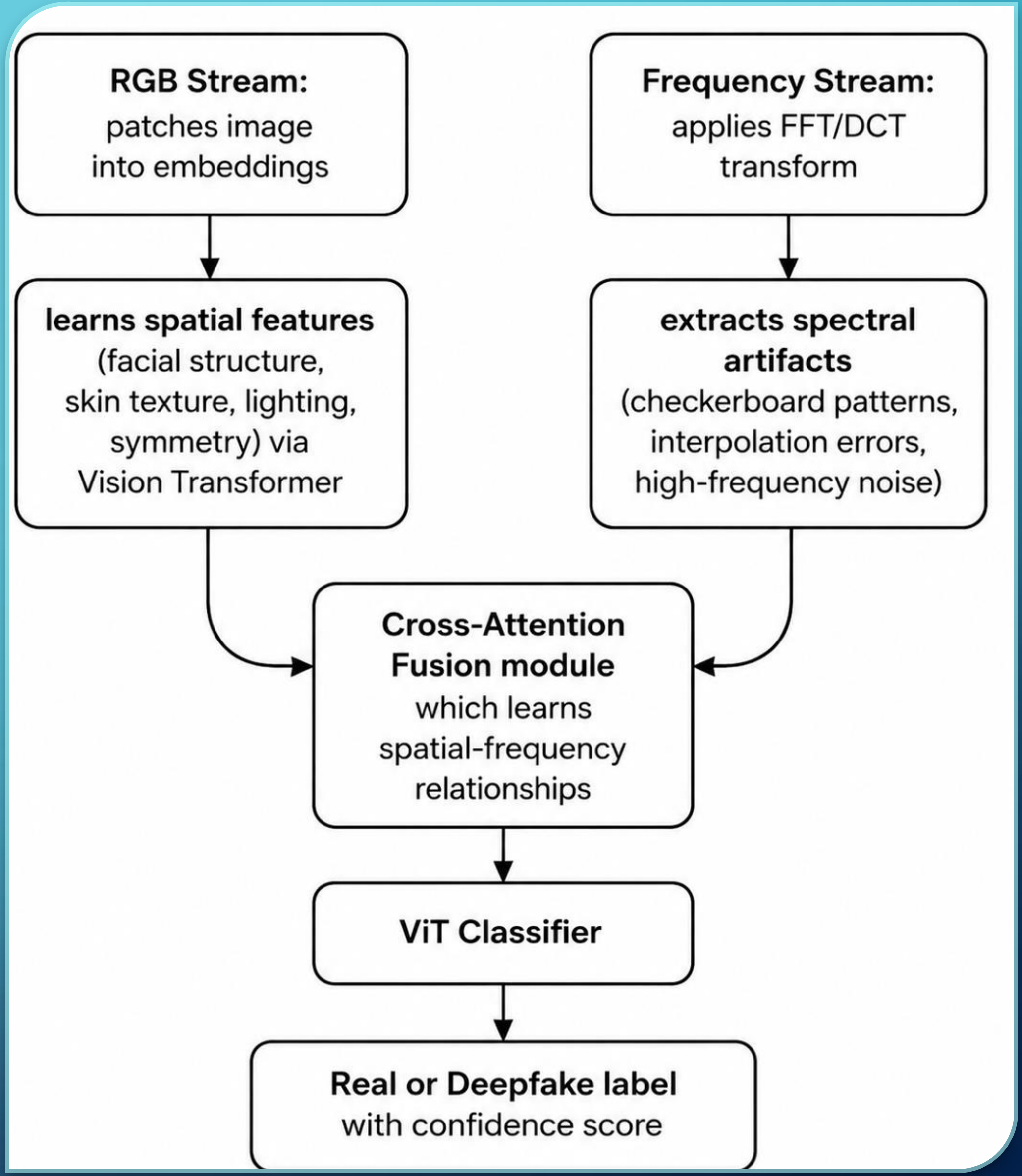
- Displays hidden artifacts using **FFT/DCT**.
- Detects **high-frequency noise** and **compression artifacts.**

**Confidence Score**

- Shows prediction confidence.
- **Example:** Deepfake (**95.32**%)

# DUAL-STREAM ARCHITECTURE & CROSS-ATTENTION FUSION

- **Why Cross-Attention Fusion?**
- Rather than simple concatenation, the cross-attention mechanism enables **meaningful interaction** between spatial facial regions and their corresponding frequency characteristics producing a richer, more discriminative representation.



# PROPOSED APPROACH VS. EXISTING SOLUTIONS

| Feature | Existing Solutions | Proposed Framework |
|---|---|---|
| **Architecture** | CNNs or ViT used independently | Explainable Dual-Stream ViT |
| **Feature Fusion** | Concatenation or independent processing | Cross-attention RGB–Frequency Fusion |
| **Frequency Analysis** | Limited or absent | FFT/DCT-based extraction |
| **Explainability** | Limited or unavailable | Attention Rollout + Frequency Saliency |
| **Output** | Prediction label only | Prediction, confidence, heatmaps, PDF report |
| **Reproducibility** | Limited (e.g., TruthScan proprietary) | Fully documented research framework |

# BASELINE PERFORMANCE

## KEY BENCHMARK INSIGHTS

ALL BASELINES ACHIEVE STRONG WITHIN-DOMAIN AUC (>0.94), BUT CROSS-DOMAIN PERFORMANCE DROPS SHARPLY — CONFIRMING **GENERALIZATION** AS THE PRIMARY UNRESOLVED CHALLENGE.

EXPECTED OUTCOME: IMPROVED CROSS-DATASET AUC VIA RGB–FREQUENCY FUSION AND TARGETED AUGMENTATION AGAINST COMPRESSION AND SCREENSHOT ATTACKS.

# EVALUATION STRATEGY

- Better cross-dataset generalization
- Improved robustness to compression and post-processing
- Higher explainability using Attention Rollout and Frequency Visualization
- Competitive inference time with baseline models

# EXPECTED OUTCOMES

- **Accurate detection** of real and AI-generated/deepfake facial images.
- **Improved cross-dataset generalization** for unseen manipulation techniques.
- **Robust performance** on compressed, low-quality, and post-processed images.
- **Enhanced explainability** through Attention Rollout and frequency-domain visualization.
- **Automated forensic PDF reports** with prediction, confidence score, and visual explanations.
- **Competitive performance** compared with baseline models (Xception, EfficientNet-B4, F3Net, UCF, and SPSL).